

Time Series Forecasting Models

Time series forecasting involves analyzing data that evolves over some period of time and then utilizing statistical models to make predictions about future patterns and trends. It takes into account the sequential nature of data where each observation is dependent on previous observations.

The significance and practical uses of time series analysis:

1. Forecasting/ Prediction:

Through the use of existing historical data, time series analysis enables the prediction of future values and trends.

2. Trend Analysis:

Time series analysis helps us find and understand patterns in data that change over time. It shows us how things are changing and how much they are changing.

3. Seasonal Patterns:

Time series analysis is a helpful tool that allows us to find and understand patterns that repeat over time. For example, in industries like retail, tourism, and agriculture, where things change depending on the season or specific times of the year, time series analysis helps us predict and plan for those changes.

4. Anomaly Detection:

Time series analysis is a useful tool that can help us find unusual things in data. It can detect when something strange or unexpected happens, like errors or unusual events. This is important because it allows us to catch fraud, keep an eye on how well a system is working, or notice when something is behaving strangely.

5. Financial Analysis:

Time series analysis is a valuable tool used in finance to study stock prices, predict market trends, manage investment portfolios, and assess risks. It helps investors and financial institutions make smarter decisions by analyzing past patterns and trends in the stock market.

6. Environmental Monitoring:

Time series analysis is used in environmental sciences to study and predict things like weather patterns, air quality, water levels, and other environmental factors.

Characteristics of time series data

1. Temporal Ordering:
Time series data is ordered chronologically, with each observation occurring after the previous one. This ordering is essential for analyzing trends and patterns.
2. Time Dependency:
In a time series, each observation is influenced by the preceding observations, creating a sequential relationship where the value at a given time depends on the values that occurred before it.
3. Irregular Sampling:
Analyzing and forecasting time series data can be challenging when there are irregular or uneven time intervals between observations. Dealing with missing or irregularly spaced data points necessitates the use of suitable techniques.

Components of time series:

1. Trend:
This represents the long-term direction or tendency of the data. It captures the overall upward or downward movement over time. Trends can be linear (constant increase or decrease) or nonlinear (curved or oscillating).
2. Seasonality:
Refers to patterns that repeat at fixed intervals within a time series. These patterns can be daily, weekly, monthly, or yearly. External factors such as weather conditions, holidays, or economic cycles often have an impact on seasonality.
3. Noise(random fluctuations/ irregularities) :
Represents the unpredictable and random variations in the data and includes factors that cannot be explained by trend or seasonality. Measurement errors, random events, or unidentified factors can contribute to the presence of noise in the data.

Overview of Time Series Models

Below is a brief overview of some commonly used time series models:

1. Moving Average (MA) Model:

This model calculates the average of past observations with the aim of predicting future values. It is useful for capturing short-term fluctuations and random variations in the data.

Assumptions: The observations are a linear combination of past error terms, and there is no autocorrelation between the error terms.

Parameters: The order of the model (q) determines the number of lagged error terms to include.

Strengths: MA models are effective in capturing short-term dependencies and smoothing out random fluctuations in the data.

2. **Autoregressive (AR) Model:** This model predicts future values based on a linear combination of past observations.

Assumptions: It assumes that the future values depend on the previous values, capturing long-term trends and dependencies.

Parameters: The order of the model (p) determines the number of lagged observations to include.

Strengths: AR models are useful for capturing long-term dependencies and trends in the data.

3. **Autoregressive Moving Average (ARMA) Model:** The ARMA model combines the AR and MA models to capture both short-term and long-term patterns in the data. It is effective for analyzing stationary time series data.

Assumptions: The observations are a linear combination of past observations and past error terms, and there is no autocorrelation between the error terms.

Parameters: The orders of the AR and MA components (p and q) determine the number of lagged observations and error terms to include.

Strengths: ARMA models combine the strengths of AR and MA models, capturing both short-term and long-term dependencies in the data.

4. **Autoregressive Integrated Moving Average (ARIMA) Model:** This model extends the ARMA model by incorporating differencing to handle non-stationary data. It is suitable for data with trends or seasonality.

Assumptions: The data is stationary after differencing, meaning the differences between consecutive observations are stationary.

Parameters: The orders of the AR, I, and MA components (p , d , and q) determine the number of lagged observations, differencing, and lagged error terms to include.

Strengths: ARIMA models can handle non-stationary data by incorporating differencing, making them suitable for time series with trends or seasonality.

5. **Seasonal ARIMA (SARIMA) Model:** This model is an extension of the ARIMA model and includes seasonal components. It is useful for analyzing and forecasting data with recurring seasonal patterns.

Assumptions: The data exhibits seasonal patterns as well as trends and dependencies.

Parameters: The orders of the seasonal AR, I, and MA components (P , D , and Q) determine the number of lagged seasonal observations, seasonal differencing, and lagged seasonal error terms to include.

Strengths: SARIMA models are effective for analyzing and forecasting time series data with seasonal patterns.

6. **Exponential Smoothing Models:** Exponential smoothing models, such as Simple Exponential Smoothing (SES) and Holt-Winters' Exponential Smoothing, use weighted averages of past observations to make predictions and are effective for capturing trends and seasonality in the data.

Assumptions: The future values are a weighted sum of past observations, with exponentially decreasing weights.

Parameters: The smoothing factor (alpha) determines the weight given to recent observations.

Strengths: Exponential smoothing models are simple yet effective for forecasting, providing good results for data with smooth trends and no seasonality.

7. **Vector Autoregression (VAR) Model:** This model is used when multiple time series variables interact with each other and it captures the relationships and dependencies between variables, making it suitable for macroeconomic forecasting.

Assumptions: The time series variables are interdependent and follow a multivariate autoregressive process.

Parameters: The orders of the VAR model (p) determine the number of lagged observations to include for each variable.

Strengths: VAR models can capture the interdependencies between multiple time series variables, making them suitable for macroeconomic forecasting and analyzing complex systems.

8. **Machine Learning Models:** Machine learning algorithms, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, can also be applied to time series analysis. These models can capture complex patterns and dependencies in the data.

Assumptions: These models can capture complex patterns and dependencies in the data without explicit assumptions about the underlying process.

Parameters: The architecture and hyperparameters of the specific machine learning model.

Strengths: Machine learning models can handle nonlinear relationships and capture long-term dependencies, making them suitable for complex time series analysis.

Metrics for evaluating time series models

1. **Mean Absolute Error (MAE):**

This metric measures the average absolute difference between the predicted and actual values. It provides a straightforward measure of the model's accuracy.

2. **Root Mean Squared Error (RMSE):**

RMSE calculates the square root of the average squared difference between the predicted and actual values. It penalizes larger errors more heavily than MAE.

3. **Mean Absolute Percentage Error (MAPE):**

MAPE calculates the average percentage difference between the predicted and actual values. It provides a relative measure of the model's accuracy.

4. **Forecast Bias:**

Forecast bias measures the tendency of the model to consistently overestimate or underestimate the actual values. A bias close to zero indicates a well-calibrated model.

Cross-validation techniques:

1. **Rolling Window Cross-Validation:**

In this approach, a fixed-size training window is used to train the model, and a fixed-size validation window is used to evaluate its performance. The window is then rolled forward in time until all data points are evaluated.

2. **Walk-Forward Validation:Walk-Forward Validation:**

This method is similar to rolling window cross-validation but involves using a sliding window that moves one step at a time. The model is trained on the available data up to a certain point and then tested on the next data point.

What is Stationarity? (Dont Move)

To have same pattern in fixed interval of time . Mean, Variance and autocorrelation depends only on difference in time, not in location.

Weak stationarity means finite variance but strong does not.

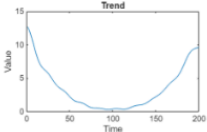
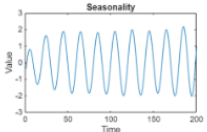
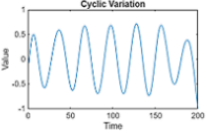
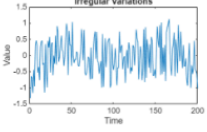
Key Properties of a Stationary Time Series:

Constant mean over time

Constant variance over time

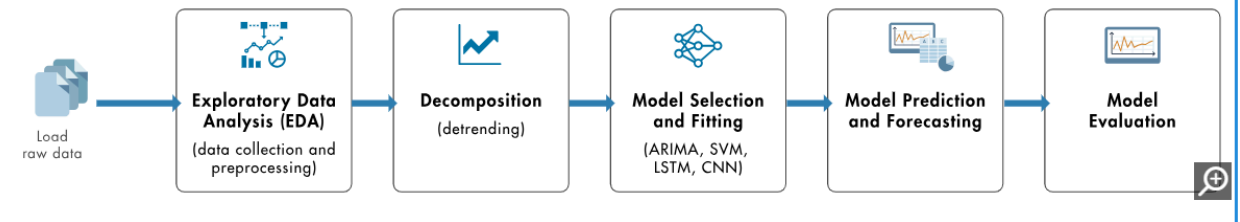
Constant autocovariance (correlation with past values) over time.
Most statistical forecasting models (like **ARIMA**) assume the input series is stationary. If it's not, the model results may be misleading.

Characteristics of Time Series Data Components

	Component	Definition	Example
	Trend	Overall direction of the data over time, such as increasing, decreasing, or constant	Adoption of EV technology, revenue for a start-up company
	Seasonality	Repeating patterns of data over a set of periods of time, such as daily, monthly, or yearly	Electricity demand peaks during summer months
	Cyclic Variations	Repeating but nonseasonal patterns in the data, occurring over several years	Real estate market cycles
	Irregular Variations	Unpredictable ups and downs that cannot be explained by other components	Sudden stock market crash due to unforeseen events

Time Series Analysis Steps

understanding, modeling, and forecasting data points collected over time.



Decomposition

Decomposition is a technique used to break down time series data into its fundamental components—trends, seasonal, cyclic, and remainder—making it easier to analyze the underlying patterns and interpret data.

Types of Time Series Analysis

Time series analyses can be classified into categories that describe, explain, forecast, and optimize data trends and patterns effectively:

Descriptive analysis summarizes the key features of a data set. It helps in understanding the underlying patterns, trends, and seasonal variations in the data.

Explanatory analysis aims to understand the relationships and causal effects within the data. It seeks to explain why certain patterns or trends occur by examining the underlying factors and correlations.

Predictive analysis uses historical data, statistical models, and machine learning algorithms to predict future trends and patterns.

Prescriptive analysis provides recommendations and actions based on predictive insights. It suggests optimal decisions by evaluating different scenarios and their potential outcomes.

Time Series Analysis of Sales Data

This portion uses different methodology to analyze time series [records](#). We can use a wonderful python module stats models that we can use to analyse the data more effectively.

Introduction to statsmodels

Statsmodels is a python module that provides classes and functions for estimation of many different statistical models, as well as for conducting statistical tests and statistical data exploration.

Time Series Stationarity

A time series is stationary if the mean and variance are fixed between two equidistance points. That is, no matter where you take your observations, the result should be same. A series that shows seasonality is not stationary.

How to test if a series is stationary?

A test for stationarity usually involves a unit root hypothesis test, where the null hypothesis is that the series is nonstationary, and contains a unit root. The alternate hypothesis supports stationarity. The augmented Dickey-Fuller and Kwiatkowski-Phillips-Schmidt-Shin tests are stationarity tests.

Augmented Dickey-Fuller Test

To determine whether a series is stationary we can use the augmented Dickey-Fuller Test. In this test the null hypothesis states that (this is also called a unit test).

The test returns several statistics we'll see in a moment. Our focus is on the p-value. A small p-value ($p < 0.05$) indicates strong evidence against the null hypothesis.

To demonstrate, we'll use a dataset we know is not stationary, the `airline_passenger` dataset. First, let's plot the data along with a 12-month rolling mean and standard deviation:

Granger Causality Tests

The Granger causality test is a hypothesis test to determine if one time series is useful in forecasting another.

While it is fairly easy to measure correlations between series - when one goes up the other goes up, and vice versa - it's another thing to observe changes in one series correlated to changes in another after a consistent amount of time.

This may indicate the presence of causality, that changes in the first series influenced the behavior of the second. However, it may also be that both series are affected by some third factor, just at different rates. Still, it can be useful if changes in one series can predict upcoming changes in another, whether there is causality or not. In this case we say that one series "Granger-causes" another.

In the case of two series, y and x , the null hypothesis is that lagged values of x do not explain variations in y .

In other words, it assumes that X_T doesn't Granger-cause Y_T .

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Time Series Forecasting General Methods

Forecasting with Holt-Winter method

This method is another name of Triple Exponential Smoothing. It takes seasonality into account. When we expect regular fluctuations in the future, this model attempts to map the seasonal behavior.

Time Series Stationarity and Differencing

Stationarity

Time series said to be stationary if it does not exhibit trends or seasonality i.e fluctuations in the data are entirely due to outside factors and noises.

Differencing

Non-stationary data can be made to look stationary through differencing. A simple differencing method calculates the difference between consecutive points.